

MATHEMATICAL POSTULATES OF QUANTUM MECHANICS

# Unit 1: Complex Numbers

The number system quantum mechanics is written in. What a complex number is, how to calculate with one by hand, and how Euler's formula turns multiplication into rotation.

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COURSE

25AID432 Mathematical Postulates of  
Quantum Mechanics

UNIT

1 of 3

SESSIONS

1 to 3

SYLLABUS

What is a complex number, doing math with complex numbers, Euler's formula and the polar form

READING TIME

About ninety minutes

OFFERING

Odd Semester 2026-27

# Before you start

This is a course about the mathematics of quantum mechanics, and it opens with complex numbers because they are the language the theory is written in, and nothing in Units 2 and 3 can be read without them.

Here is the reason in three sentences. In quantum mechanics the state of a system is described by a list of numbers called **amplitudes**, and those amplitudes are complex. When two amplitudes add, the result depends on the *angle* each one points in, and that angle (the **phase**) is what produces interference: two contributions can reinforce each other or cancel completely. A single real number has no angle to speak of, so a theory built on real numbers alone cannot describe this.

You will meet one more fact now and prove nothing about it until Unit 3. If  $\alpha$  is an amplitude, then  $|\alpha|^2$ , the square of its size, is a **probability**. That is all the physics this unit needs. Everything else here is mathematics you can check with pencil and paper.

The unit follows the syllabus in three sessions. Session 1 covers what a complex number is and how to draw one. Session 2 covers arithmetic: adding, multiplying, conjugating, taking the modulus and dividing. Session 3 covers Euler's formula, the polar form and the idea of phase, and ends with a short look at why the phase matters in quantum mechanics.

## BY THE END OF THIS UNIT YOU WILL BE ABLE TO

- Write a complex number as  $a + bi$ , name its real and imaginary parts, and plot it in the complex plane.
- Add, subtract, multiply and divide complex numbers by hand, and simplify any power of  $i$ .
- Compute the complex conjugate and the modulus, and use  $zz^* = |z|^2$ .
- Convert between Cartesian form  $a + bi$  and polar form  $re^{i\theta}$ , choosing the correct quadrant for the angle.
- State Euler's formula and use it to multiply, divide and raise complex numbers to powers.
- Explain why  $|z|^2$  does not depend on the phase of  $z$ , while  $|z_1 + z_2|^2$  depends on the difference of the two phases.
- Check any of these hand calculations in Python with NumPy.

## KEY TERMS IN THIS UNIT

imaginary unit · complex number · real part ·  
imaginary part · complex plane · Argand diagram ·  
complex conjugate · modulus · argument · phase ·

Cartesian form · polar form · Euler's formula · unit  
circle · principal argument · roots of unity · amplitude  
· relative phase · global phase · interference

# What a complex number is

## The imaginary unit

No real number squares to  $-1$ : a positive number times itself is positive, and so is a negative number times itself. So the equation  $x^2 = -1$  has no real solution.

Mathematicians fixed this by adding one new number to the system and giving it a name.

### DEFINITION · THE IMAGINARY UNIT

The **imaginary unit**  $i$  is a number with the property

$$i^2 = -1.$$

Nothing else is assumed about it. Every rule of ordinary algebra (brackets, the order of multiplication, collecting like terms) still holds, and whenever  $i^2$  appears you replace it by  $-1$ .

The name “imaginary” is a historical accident from the 1600s, when these numbers were treated with suspicion. The name stuck; the suspicion should not. A complex number is as well defined as a negative number or a fraction, and you will see in a few pages that it has a plain geometric meaning.

## Real part, imaginary part and equality

### DEFINITION · COMPLEX NUMBER

A **complex number** is a number of the form

$$z = a + bi, \quad a, b \in \mathbb{R}.$$

The real number  $a$  is the **real part**, written  $\operatorname{Re}(z)$ . The real number  $b$  is the **imaginary part**, written  $\operatorname{Im}(z)$ . The set of all complex numbers is written  $\mathbb{C}$ .

Notice that the imaginary part is  $b$ , a real number, and not  $bi$ . For  $z = 4 - 3i$  we have  $\operatorname{Re}(z) = 4$  and  $\operatorname{Im}(z) = -3$ .

Two complex numbers are **equal** when their real parts are equal *and* their imaginary parts are equal. So one equation between complex numbers is really two equations between real numbers. This is used constantly.

### WORKED EXAMPLE · ONE COMPLEX EQUATION, TWO REAL EQUATIONS

Find real numbers  $x$  and  $y$  with  $(x + y) + (x - y)i = 5 + i$ .

Match real parts:  $x + y = 5$ . Match imaginary parts:  $x - y = 1$ .

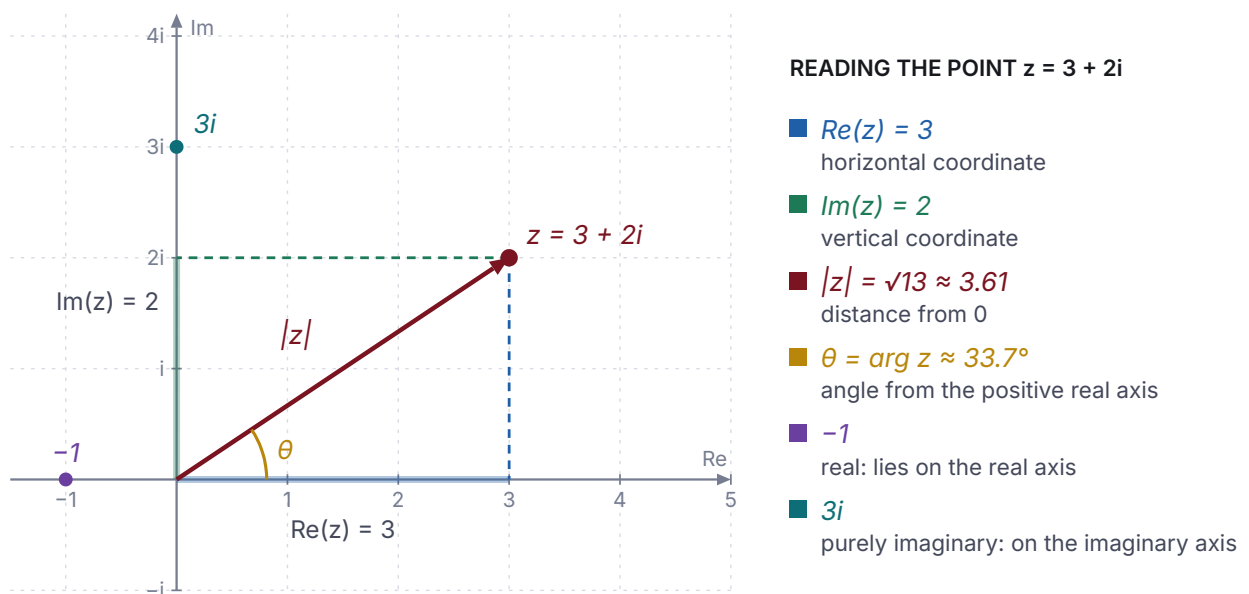
Adding the two equations gives  $2x = 6$ , so  $x = 3$ , and then  $y = 2$ .

### PITFALL · ENGINEERS WRITE $j$

Electrical engineers use  $i$  for current, so they write the imaginary unit as  $j$ . Python follows them: the imaginary unit is written `1j`, as in `3 + 4j`. A bare `j` is just a variable name, so `3 + 4*j` fails unless you have defined `j` yourself. In this course, on paper, it is always  $i$ .

## The complex plane

A complex number  $a + bi$  is fixed by two real numbers,  $a$  and  $b$ . So it can be drawn as the point  $(a, b)$  in a plane. The horizontal axis carries the real part and is called the **real axis**. The vertical axis carries the imaginary part and is called the **imaginary axis**. This picture is the **complex plane**, also called the **Argand diagram**.



**Figure 1.1** The complex plane. The number  $z = 3 + 2i$  is the point three units right and two units up. Its distance from the origin is the modulus  $|z|$  and its angle from the positive real axis is the argument  $\theta$ , both defined in the next two sections. Real numbers such as  $-1$  sit on the horizontal axis; purely imaginary numbers such as  $3i$  sit on the vertical axis.

You can think of  $z$  either as the point or as the arrow from the origin to that point. The arrow view is the useful one for the rest of this unit, because arrows have a length and a direction, and both will turn out to mean something.

### WORKED EXAMPLE · READING AND PLOTTING

For each number, give the real and imaginary parts and the quadrant.

| NUMBER          | REAL PART | IMAGINARY PART | WHERE IT SITS                  |
|-----------------|-----------|----------------|--------------------------------|
| $z_1 = -2 + 5i$ | -2        | 5              | second quadrant (left and up)  |
| $z_2 = 7$       | 7         | 0              | positive real axis             |
| $z_3 = -4i$     | 0         | -4             | negative imaginary axis        |
| $z_4 = -1 - i$  | -1        | -1             | third quadrant (left and down) |

## Real numbers are complex numbers too

Any real number  $a$  can be written  $a + 0i$ , so the real numbers are exactly the complex numbers whose imaginary part is zero. They form the real axis of the plane. A number with real part zero, such as  $3i$ , is called **purely imaginary**.

Nothing you know about real numbers is lost. Complex arithmetic, restricted to the real axis, is ordinary arithmetic.

### COMMON MISCONCEPTION · ONE COMPLEX NUMBER CAN BE BIGGER THAN ANOTHER

For real numbers, " $x < y$ " means  $x$  is to the left of  $y$  on the number line. There is no such ordering of the plane that is compatible with arithmetic, so statements such as  $i > 0$  or  $2 + i < 3$  have no meaning. What you *can* compare is the size of two complex numbers,  $|z|$  and  $|w|$ , which are real. That is the next section.

# Doing mathematics with complex numbers

Session 2 is arithmetic. There is one rule to remember,  $i^2 = -1$ , and every calculation below is ordinary algebra with that substitution.

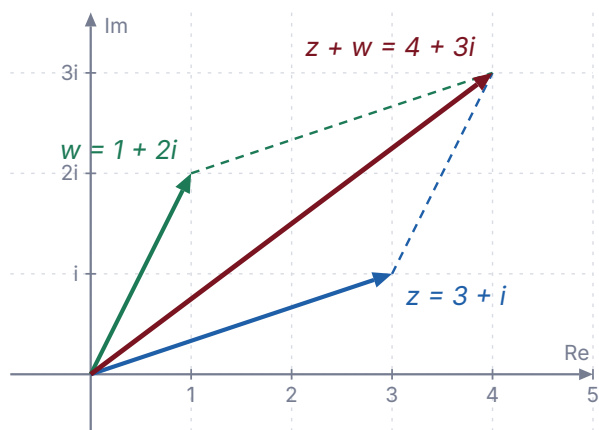
## Addition and subtraction

Add the real parts and add the imaginary parts:

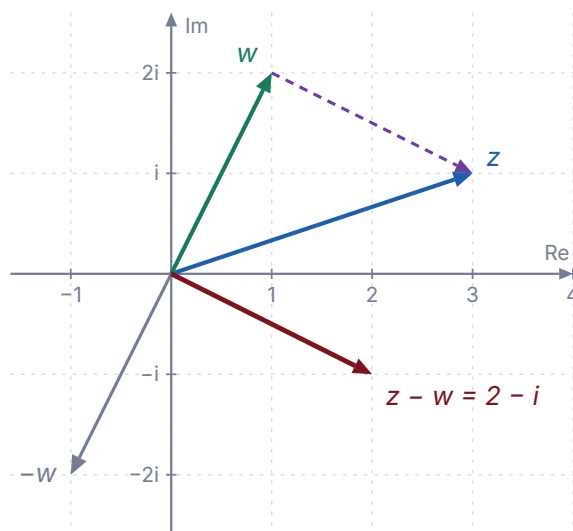
$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

Subtraction works the same way. In the plane this is exactly vector addition: place the arrow for  $w$  at the head of the arrow for  $z$ , and  $z + w$  is the arrow from the origin to where you end up. Subtracting  $w$  is adding  $-w$ , the arrow of the same length pointing the opposite way.

(a) ADD: PLACE THE ARROWS HEAD TO TAIL



(b) SUBTRACT: ADD  $-w$



**Figure 1.2** (a)  $z + w$  is the diagonal of the parallelogram built on  $z$  and  $w$ . (b)  $z - w$  is  $z + (-w)$ . It is also the arrow from the tip of  $w$  to the tip of  $z$  (dashed), moved so that it starts at the origin.

### WORKED EXAMPLE · ADDING AND SUBTRACTING

Take  $z = 3 + i$  and  $w = 1 + 2i$ , the numbers in Figure 1.2.

$$z + w = (3 + 1) + (1 + 2)i = 4 + 3i, \quad z - w = (3 - 1) + (1 - 2)i = 2 - i.$$

## Multiplication

Expand the brackets as you would for any two binomials, then use  $i^2 = -1$ :

$$(a + bi)(c + di) = ac + adi + bci + bd i^2 = (ac - bd) + (ad + bc)i.$$

Do not memorise the final formula. Expand and substitute every time; it is faster and harder to get wrong.

#### WORKED EXAMPLE · A PRODUCT

Compute  $(2 + 3i)(1 - 4i)$ .

$$(2 + 3i)(1 - 4i) = 2 - 8i + 3i - 12i^2 = 2 - 5i + 12 = 14 - 5i.$$

The term  $-12i^2$  became  $+12$ . Losing that sign is the most common slip in this whole unit.

Multiplying by a real number  $k$  scales both parts:  $k(a + bi) = ka + kbi$ . In the plane that stretches the arrow by  $k$  without turning it (or reverses it, if  $k$  is negative). Multiplying by a complex number does something more interesting, which Session 3 explains.

## Powers of $i$

Keep multiplying by  $i$ :

$$i^1 = i, \quad i^2 = -1, \quad i^3 = i^2 \cdot i = -i, \quad i^4 = (i^2)^2 = 1, \quad i^5 = i^4 \cdot i = i, \dots$$

The powers repeat with period 4. To simplify  $i^n$ , divide  $n$  by 4 and keep the remainder:  $i^n = i^{n \bmod 4}$ .

#### WORKED EXAMPLE · HIGH AND NEGATIVE POWERS

For  $i^{27}$ , write  $27 = 4 \times 6 + 3$ . Then  $i^{27} = (i^4)^6 i^3 = -i$ .

For  $i^{-1}$ , which is  $1/i$ , multiply top and bottom by  $i$ :  $\frac{1}{i} = \frac{i}{i^2} = -i$ . Check:  $i \cdot (-i) = -i^2 = 1$ .

So  $\frac{1}{i} = -i$ , a fact that turns up often enough to be worth remembering.

#### PITFALL · THE SQUARE-ROOT TRAP

It is tempting to write  $i = \sqrt{-1}$  and then

$$-1 = i \cdot i = \sqrt{-1} \sqrt{-1} = \sqrt{(-1)(-1)} = \sqrt{1} = 1.$$

The error is the third step. The rule  $\sqrt{a}\sqrt{b} = \sqrt{ab}$  is true when  $a$  and  $b$  are non-negative real numbers, and it fails for negative ones. Define  $i$  by  $i^2 = -1$  and avoid writing  $\sqrt{-1}$  at all.

## The complex conjugate

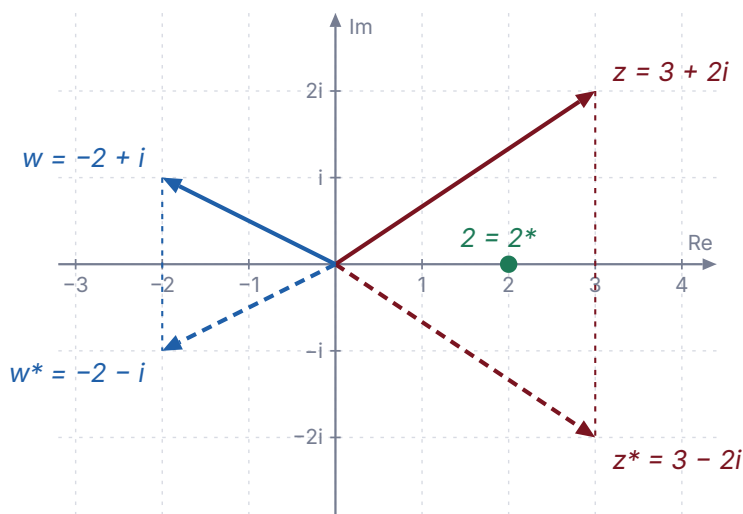
### DEFINITION · COMPLEX CONJUGATE

The **complex conjugate** of  $z = a + bi$  is

$$z^* = a - bi.$$

Keep the real part, flip the sign of the imaginary part. Mathematics books usually write  $\bar{z}$ ; physics books, and this course, write  $z^*$ .

In the plane, conjugation is reflection in the real axis. A real number lies on that axis, so it is its own conjugate.



### CONJUGATION FLIPS THE SIGN OF Im

The real axis is a mirror:  $z^*$  is the reflection of  $z$  in it.

Same distance from 0:  $|z^*| = |z|$ .  
Opposite angle:  $\arg z^* = -\arg z$ .

A point on the real axis is its own reflection, so a real number equals its conjugate.

**Figure 1.3** The conjugate  $z^*$  is the mirror image of  $z$  in the real axis. The distance from the origin is unchanged and the angle changes sign. The point 2 on the real axis is its own mirror image.

The conjugate has four properties you will use in every unit of this course. Each one follows from writing  $z = a + bi$  and  $w = c + di$  and expanding.

1. It passes through sums:  $(z + w)^* = z^* + w^*$ .
2. It passes through products:  $(zw)^* = z^*w^*$ .
3. Adding a number to its conjugate cancels the imaginary part:  $z + z^* = 2a = 2\operatorname{Re}(z)$ .  
Subtracting cancels the real part:  $z - z^* = 2bi$ .
4. Multiplying a number by its conjugate gives a real, non-negative number:

$$zz^* = (a + bi)(a - bi) = a^2 - abi + abi - b^2i^2 = a^2 + b^2.$$

Property 2 is worth checking once by hand. With  $zw = (ac - bd) + (ad + bc)i$ , the conjugate is  $(ac - bd) - (ad + bc)i$ . And  $z^*w^* = (a - bi)(c - di) = ac - adi - bci + bd i^2 = (ac - bd) - (ad + bc)i$ . The two agree.

Property 4 is the important one. Multiplying by the conjugate is how you turn a complex number into a real one, and it is the reason the conjugate appears in every formula for a probability later in the course.

## The modulus

### DEFINITION · MODULUS

The **modulus** (or absolute value) of  $z = a + bi$  is

$$|z| = \sqrt{a^2 + b^2} = \sqrt{zz^*}.$$

It is the distance from the origin to the point  $z$ , by Pythagoras. It is a real number, and only  $z = 0$  has modulus zero.

Squaring gives the form you will use most often:

$$|z|^2 = zz^* = a^2 + b^2.$$

For a real number  $a$ ,  $|a| = \sqrt{a^2}$  is the usual absolute value, so the modulus extends an idea you already know. The modulus also measures distance between two numbers:  $|z - w|$  is the length of the dashed arrow in Figure 1.2(b), which is the distance between the points  $z$  and  $w$ .

### WORKED EXAMPLE · CONJUGATE AND MODULUS

Let  $z = 3 + 4i$ . Then  $z^* = 3 - 4i$  and

$$zz^* = (3 + 4i)(3 - 4i) = 9 - 12i + 12i - 16i^2 = 9 + 16 = 25,$$

so  $|z| = \sqrt{25} = 5$ . The point  $(3, 4)$  is at distance 5 from the origin, the familiar 3-4-5 triangle.

### PITFALL · THREE WAYS TO GET THE MODULUS WRONG

**First:**  $|a + bi| \neq a + b$ . For  $3 + 4i$  that would give 7; the answer is 5.

**Second:**  $|a + bi| \neq \sqrt{a^2 + (bi)^2}$ . That gives  $\sqrt{9 - 16}$ , which is nonsense. The imaginary *part* is  $b = 4$ , a real number, and it is  $b$  that gets squared.

**Third:**  $|z|^2 \neq z^2$ . Here  $z^2 = (3 + 4i)^2 = 9 + 24i + 16i^2 = -7 + 24i$ , which is not even real. The modulus squared is  $z$  times its *conjugate*.

## Division

To divide by  $w$ , multiply the top and the bottom by  $w^*$ . The bottom becomes  $ww^* = |w|^2$ , a real number, and dividing by a real number is easy:

$$\frac{z}{w} = \frac{z w^*}{w w^*} = \frac{z w^*}{|w|^2}.$$

This is the same move as rationalising a denominator such as  $1/(1 + \sqrt{2})$  in school, where you multiply by  $1 - \sqrt{2}$  to clear the root.

#### WORKED EXAMPLE · A QUOTIENT

Compute  $\frac{1 + 2i}{3 - i}$ .

The conjugate of the bottom is  $3 + i$ , and  $(3 - i)(3 + i) = 9 + 1 = 10$ . So

$$\frac{1 + 2i}{3 - i} = \frac{(1 + 2i)(3 + i)}{10} = \frac{3 + i + 6i + 2i^2}{10} = \frac{1 + 7i}{10} = 0.1 + 0.7i.$$

Check by multiplying back:  $(0.1 + 0.7i)(3 - i) = 0.3 - 0.1i + 2.1i - 0.7i^2 = 1 + 2i$ .

One more property follows from  $zz^* = |z|^2$  and the product rule for conjugates. For any  $z$  and  $w$ ,

$$|zw|^2 = (zw)(zw)^* = zwz^*w^* = (zz^*)(ww^*) = |z|^2|w|^2,$$

so  $|zw| = |z||w|$ . **The size of a product is the product of the sizes.** Keep this in mind; Session 3 explains it geometrically.

#### NOTE · CHECK IT IN NUMPY

Python has complex numbers built in; NumPy adds functions that work on whole arrays of them. Do the calculation by hand first, then check it.

```
:::python
import numpy as np
z = 3 + 4j                                # the imaginary unit is 1j in Python
w = 1 - 2j
print(z + w, z * w, z / w)               # (4+2j) (11-2j) (-1+2j)
print(z.real, z.imag)                   # 3.0 4.0
print(np.conj(z), np.abs(z))             # (3-4j) 5.0
print(z * np.conj(z))                   # (25+0j): real, as promised
print((1 + 2j) / (3 - 1j))               # (0.1+0.7000000000000001j)
```

The last line prints `0.7000000000000001`. That is floating-point rounding, the computer storing 0.7 in binary. Your hand answer  $0.1 + 0.7i$  is exact.

# Euler's formula and the polar form

Session 3 describes a complex number by its length and its direction, and this makes multiplication, powers and phase easy to see.

Angles in this unit are in **radians**:  $\pi$  radians is  $180^\circ$ , so  $\pi/2 = 90^\circ$ ,  $\pi/3 = 60^\circ$ ,  $\pi/4 = 45^\circ$  and  $\pi/6 = 30^\circ$ . NumPy's trigonometric functions also use radians.

## Length and angle

Look again at Figure 1.1. The point  $z = a + bi$  is at distance  $r = |z|$  from the origin, and the arrow makes an angle  $\theta$  with the positive real axis. From the right-angled triangle,

$$a = r \cos \theta, \quad b = r \sin \theta,$$

so

$$z = r \cos \theta + i r \sin \theta = r(\cos \theta + i \sin \theta).$$

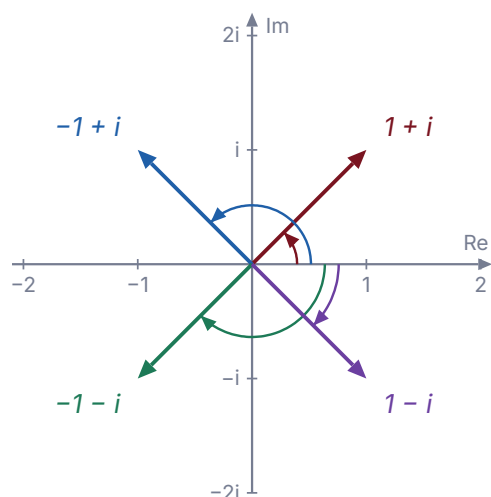
### DEFINITION · ARGUMENT

The angle  $\theta$ , measured anticlockwise from the positive real axis, is the **argument** of  $z$ , written  $\arg z$ . In physics it is also called the **phase**. The argument of 0 is not defined.

Adding  $2\pi$  to an angle brings you back to the same direction, so  $\theta$  and  $\theta + 2\pi$  describe the same number. To have one definite answer we use the **principal argument**, the value in the range  $-\pi < \theta \leq \pi$ . That is also what NumPy returns.

## Finding the angle: mind the quadrant

Going from  $(a, b)$  to  $(r, \theta)$ , the modulus is easy:  $r = \sqrt{a^2 + b^2}$ . The angle needs care. Since  $\tan \theta = b/a$ , it is tempting to write  $\theta = \arctan(b/a)$ . That is correct only when  $a > 0$ .



#### THE SAME RATIO $b/a$ , FOUR DIFFERENT ANGLES

| $z$      | $b/a$ | $\arctan(b/a)$ | true arg $z$ |
|----------|-------|----------------|--------------|
| $1 + i$  | 1     | $45^\circ$     | $45^\circ$   |
| $-1 + i$ | -1    | $-45^\circ$    | $135^\circ$  |
| $-1 - i$ | 1     | $45^\circ$     | $-135^\circ$ |
| $1 - i$  | -1    | $-45^\circ$    | $-45^\circ$  |

$\arctan$  only returns angles between  $-90^\circ$  and  $90^\circ$ , so it is wrong for every point with  $\text{Re}(z) < 0$  (red).

**Figure 1.4** The points  $1 + i$  and  $-1 - i$  have the same ratio  $b/a = 1$ , so  $\arctan(b/a)$  gives  $45^\circ$  for both. They point in opposite directions. The ratio throws away the signs of  $a$  and  $b$  separately, and those signs are what fix the quadrant.

#### PITFALL · ARCTAN IS THE WRONG QUADRANT HALF THE TIME

The calculator's  $\arctan$  only returns angles between  $-90^\circ$  and  $90^\circ$ , which is the right half of the plane. For any point with  $a < 0$  it is out by  $180^\circ$ . The safe method: **draw the point first**, decide the quadrant, find the reference angle  $\arctan |b/a|$ , then place it in that quadrant. On a computer, use the two-argument function `np.arctan2(b, a)` or simply `np.angle(z)`, which look at the signs of  $a$  and  $b$  separately.

#### WORKED EXAMPLE · CARTESIAN TO POLAR, SECOND QUADRANT

Find  $r$  and  $\theta$  for  $z = -1 + \sqrt{3}i$ .

The modulus is  $r = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{4} = 2$ .

The point is left of the imaginary axis and above the real axis: second quadrant. The reference angle is  $\arctan(\sqrt{3}/1) = \pi/3$ . In the second quadrant the angle is  $\pi - \pi/3 = 2\pi/3$ , which is  $120^\circ$ .

Blindly typing  $\arctan(\sqrt{3}/(-1)) = \arctan(-\sqrt{3})$  gives  $-\pi/3$ , a point in the fourth quadrant. Wrong by exactly  $\pi$ .

Check:  $2(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}) = 2(-\frac{1}{2} + i \frac{\sqrt{3}}{2}) = -1 + \sqrt{3}i$ .

## Euler's formula

Now take the numbers with modulus 1. They all sit on the **unit circle**, the circle of radius 1 around the origin, and each one has the form  $\cos \theta + i \sin \theta$  for some angle  $\theta$ . Give this number a temporary name,  $E(\theta) = \cos \theta + i \sin \theta$ , and multiply two of them:

$$E(\alpha) E(\beta) = (\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta).$$

The two brackets are the compound-angle formulas from school trigonometry, so

$$E(\alpha) E(\beta) = \cos(\alpha + \beta) + i \sin(\alpha + \beta) = E(\alpha + \beta).$$

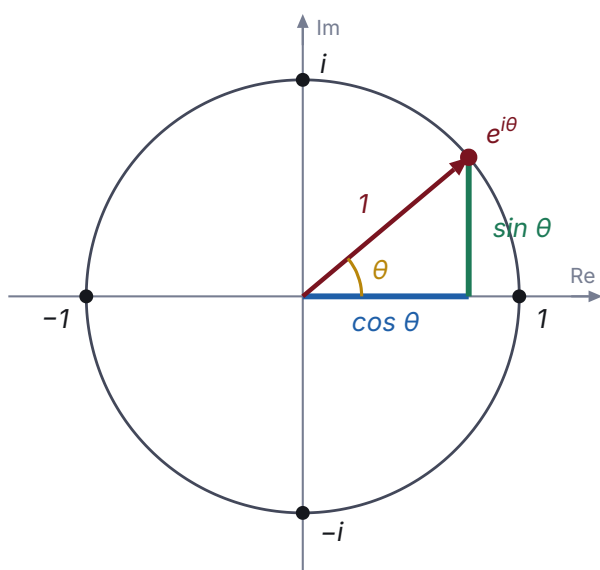
Multiplying two of these numbers *adds their angles*. Also  $E(0) = 1$ . These are exactly the rules of an exponential:  $x^\alpha x^\beta = x^{\alpha+\beta}$  and  $x^0 = 1$ . Leonhard Euler showed that  $E(\theta)$  is in fact the exponential function evaluated at  $i\theta$ .

#### DEFINITION · EULER'S FORMULA

For every real  $\theta$ ,

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

The number  $e^{i\theta}$  is the point on the unit circle at angle  $\theta$ .



#### EULER'S FORMULA ON THE UNIT CIRCLE

$$e^{i\theta} = \cos \theta + i \sin \theta$$

The point at angle  $\theta$  on the circle of radius 1 has coordinates  $(\cos \theta, \sin \theta)$ .

Its distance from 0 is always 1:  
 $\cos^2 \theta + \sin^2 \theta = 1$ .

$\theta = 0, \pi/2, \pi, 3\pi/2$  land on 1,  $i$ ,  $-1$ ,  $-i$ .  
 $\theta$  and  $\theta + 2\pi$  give the same point.

**Figure 1.5** The number  $e^{i\theta}$  is the point at angle  $\theta$  on the unit circle. Its real part is  $\cos \theta$  and its imaginary part is  $\sin \theta$ . The four marked points are  $e^{i0} = 1$ ,  $e^{i\pi/2} = i$ ,  $e^{i\pi} = -1$  and  $e^{i3\pi/2} = -i$ .

Several facts can be read straight off the circle.

- $|e^{i\theta}| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$  for every  $\theta$ . Multiplying by  $e^{i\theta}$  never changes a modulus.
- $e^{i\pi/2} = i$ ,  $e^{i\pi} = -1$  and  $e^{i2\pi} = 1$ . The last says that a full turn brings you back to the start.
- Rearranged,  $e^{i\pi} = -1$  becomes  $e^{i\pi} + 1 = 0$ , which connects  $e$ ,  $i$ ,  $\pi$ , 1 and 0 in one line and is often called Euler's identity.
- $e^{i(\theta+2\pi)} = e^{i\theta}$ . Angles that differ by a full turn give the same number.

### BEYOND THE SYLLABUS · WHERE THE FORMULA REALLY COMES FROM

The argument above shows that  $\cos \theta + i \sin \theta$  behaves like an exponential. The full proof uses power series, which you will meet in a calculus course. The exponential function can be written as the infinite sum

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Put  $x = i\theta$  and use  $i^2 = -1$ ,  $i^3 = -i$ ,  $i^4 = 1$ . The terms with even powers are real and the odd ones are imaginary:

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right).$$

The two brackets are exactly the power series of  $\cos \theta$  and  $\sin \theta$ . You are not expected to reproduce this; it is here so that you know the formula is a proved theorem.

## The polar form

Combine  $z = r(\cos \theta + i \sin \theta)$  with Euler's formula.

### DEFINITION · POLAR FORM

Every non-zero complex number can be written

$$z = re^{i\theta}, \quad r = |z| > 0, \quad \theta = \arg z.$$

This is the **polar form**. The form  $a + bi$  is the **Cartesian form**. The conversions are

$$a = r \cos \theta, \quad b = r \sin \theta, \quad r = \sqrt{a^2 + b^2}, \quad \theta = \text{angle of } (a, b) \text{ in the correct quadrant.}$$

### WORKED EXAMPLE · POLAR TO CARTESIAN

Write  $4e^{i\pi/6}$  and  $\sqrt{2}e^{-i\pi/4}$  in the form  $a + bi$ .

$$4e^{i\pi/6} = 4\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) = 4\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right) = 2\sqrt{3} + 2i.$$

For the second,  $\cos(-\pi/4) = \frac{1}{\sqrt{2}}$  and  $\sin(-\pi/4) = -\frac{1}{\sqrt{2}}$ , so

$$\sqrt{2}e^{-i\pi/4} = \sqrt{2}\left(\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}\right) = 1 - i.$$

A negative angle means clockwise from the positive real axis, which puts  $1 - i$  in the fourth quadrant, as it should be.

## Multiplication is rotation and scaling

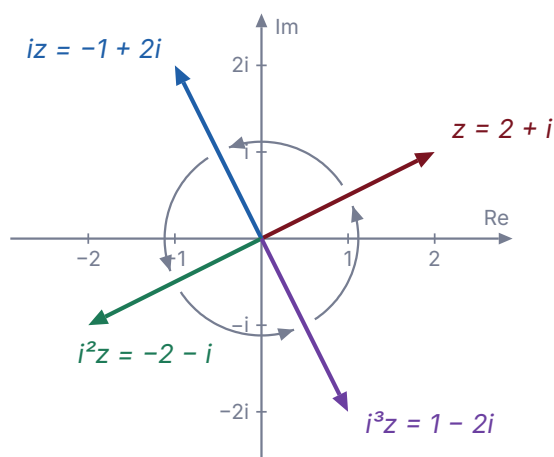
Here is the payoff. Take  $z = re^{i\theta}$  and  $w = se^{i\phi}$ . The ordinary rules for exponents give

$$zw = re^{i\theta} se^{i\phi} = (rs)e^{i(\theta+\phi)}.$$

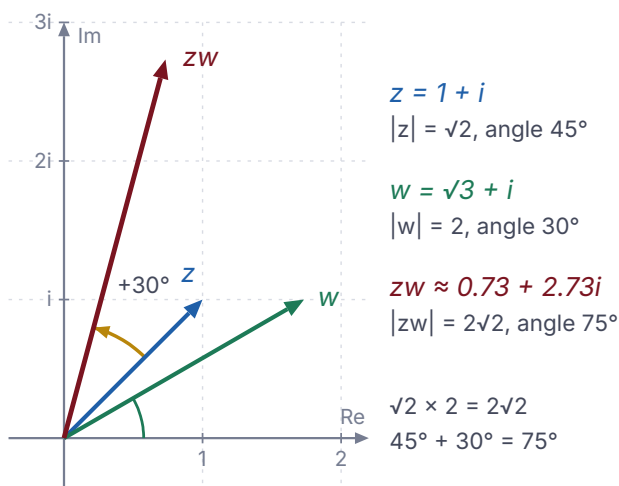
**Multiply the moduli, add the angles.** In the plane, multiplying  $z$  by  $w$  stretches the arrow of  $z$  by the factor  $|w|$  and turns it anticlockwise by the angle  $\arg w$ . This is the geometric reason behind  $|zw| = |z||w|$  from Session 2.

The special case  $w = i = e^{i\pi/2}$  is worth seeing on its own: multiplying by  $i$  turns any arrow by  $90^\circ$  anticlockwise and leaves its length alone. Doing it twice turns by  $180^\circ$ , which is multiplication by  $-1$ . That is what  $i^2 = -1$  means geometrically.

(a) MULTIPLY BY  $i$ : TURN  $90^\circ$  ANTICLOCKWISE



(b) MODULI MULTIPLY, ANGLES ADD



**Figure 1.6** (a) Each multiplication by  $i$  is a quarter turn anticlockwise; four of them return  $z$  to itself because  $i^4 = 1$ . (b) Multiplying  $z = 1 + i$  by  $w = \sqrt{3} + i$  stretches  $z$  by  $|w| = 2$  and turns it by  $\arg w = 30^\circ$ .

#### WORKED EXAMPLE · A PRODUCT IN POLAR FORM

Multiply  $z = 1 + i$  and  $w = \sqrt{3} + i$  both ways.

**Polar.**  $|z| = \sqrt{2}$  and  $\arg z = \pi/4$ .  $|w| = \sqrt{3+1} = 2$  and  $\arg w = \pi/6$  (reference angle  $\arctan(1/\sqrt{3})$ , first quadrant). So

$$zw = \sqrt{2}e^{i\pi/4} \cdot 2e^{i\pi/6} = 2\sqrt{2}e^{i5\pi/12},$$

a modulus of  $2\sqrt{2} \approx 2.83$  at an angle of  $75^\circ$ .

**Cartesian.**  $(1+i)(\sqrt{3}+i) = \sqrt{3}+i+\sqrt{3}i+i^2 = (\sqrt{3}-1) + (\sqrt{3}+1)i \approx 0.732 + 2.732i$ .

Both describe the same number. As a bonus, comparing real parts gives  $2\sqrt{2}\cos 75^\circ = \sqrt{3}-1$ , so  $\cos 75^\circ = \frac{\sqrt{3}-1}{2\sqrt{2}}$ , a result that takes more work by trigonometry alone.

## Division, powers and the conjugate in polar form

The same exponent rules handle the other operations.

$$\frac{z}{w} = \frac{re^{i\theta}}{se^{i\phi}} = \frac{r}{s}e^{i(\theta-\phi)}, \quad z^n = r^n e^{in\theta}, \quad z^* = re^{-i\theta}.$$

Division divides the moduli and subtracts the angles. A power raises the modulus to that power and multiplies the angle. The conjugate keeps the modulus and flips the sign of the angle, which is the reflection of Figure 1.3. With it you can check  $zz^* = re^{i\theta} re^{-i\theta} = r^2 e^{i0} = r^2$ , the polar version of  $zz^* = |z|^2$ .

#### WORKED EXAMPLE · A POWER THE EASY WAY

Compute  $(1 + i)^8$ .

Expanding eight brackets is slow. In polar form  $1 + i = \sqrt{2} e^{i\pi/4}$ , so

$$(1 + i)^8 = (\sqrt{2})^8 e^{i8\pi/4} = 16 e^{i2\pi} = 16.$$

Check the first step by hand:  $(1 + i)^2 = 1 + 2i + i^2 = 2i$ , so  $(1 + i)^8 = (2i)^4 = 16i^4 = 16$ .

#### PITFALL · ADDITION DOES NOT WORK IN POLAR FORM

The sum  $re^{i\theta} + se^{i\phi}$  is *not*  $(r + s)e^{i(\theta+\phi)}$ , and there is no neat polar formula for a sum. To add, convert to Cartesian form, add, and convert back if you need to. Multiply in polar, add in Cartesian.

#### NOTE · CHECK IT IN NUMPY

The polar-form examples above, checked. Compare each printed value with your hand answer.

```
:::python
import numpy as np
z = -1 + np.sqrt(3) * 1j
print(np.abs(z))           # 2.0
print(np.angle(z))         # 2.0943951023931957, i.e. 2π/3
print(np.degrees(np.angle(z))) # 120.00000000000001
print(np.arctan(z.imag / z.real)) # -1.047...: wrong quadrant
print(np.exp(1j * np.pi / 6))  # (0.866...+0.49999999999999994j)
print(np.exp(1j * np.pi))      # (-1+1.2246467991473532e-16j)
w = (1 + 1j) * (np.sqrt(3) + 1j)
print(np.abs(w), np.degrees(np.angle(w))) # 2.82842712474619 75.0
```

`np.exp(1j * np.pi)` prints  $-1$  plus a tiny imaginary part, about  $10^{-16}$ . The computer's  $\pi$  is a rounded value, so the result is  $-1$  to within rounding. Treat anything that small as zero.

#### BEYOND THE SYLLABUS · ROOTS OF UNITY

Which numbers  $z$  satisfy  $z^n = 1$ ? Write  $z = re^{i\theta}$ . Then  $z^n = r^n e^{in\theta}$  must equal  $1 = 1 \cdot e^{i2\pi k}$  for some whole number  $k$ . So  $r = 1$  and  $n\theta = 2\pi k$ , giving the  $n$  solutions

$$z = e^{i2\pi k/n}, \quad k = 0, 1, \dots, n-1.$$

They are equally spaced around the unit circle. For  $n = 4$  they are  $1, i, -1, -i$ . For  $n = 3$  they are  $1$  and  $-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$ . The same idea reappears in the quantum Fourier transform, if you go on to a quantum computing course.

# Why this matters for quantum mechanics

## Phase

When you write  $z = re^{i\theta}$ , the factor  $r$  says how big  $z$  is and the factor  $e^{i\theta}$  says which way it points. The second factor, a number of modulus 1, is called a **phase factor**, and  $\theta$  is the phase. Multiplying  $z$  by a phase factor  $e^{i\gamma}$  rotates it by  $\gamma$  and leaves the modulus alone:

$$|e^{i\gamma}z| = |e^{i\gamma}| |z| = |z|.$$

Every probability in quantum mechanics is the modulus squared of an amplitude, so this line already says something physical.

## Amplitudes and interference

In Unit 3 the state of a qubit will be described by two complex numbers,  $\alpha$  and  $\beta$ , called amplitudes, and the probabilities of the two possible measurement outcomes will be  $|\alpha|^2$  and  $|\beta|^2$ . You do not need to know what a qubit is yet. What you can already see is what this unit's mathematics says about such a rule.

**A single amplitude hides its phase.** If  $\alpha = re^{i\theta}$ , then

$$|\alpha|^2 = \alpha\alpha^* = re^{i\theta}re^{-i\theta} = r^2.$$

The phase  $\theta$  has disappeared. The amplitudes  $\frac{1}{2}$ ,  $\frac{i}{2}$  and  $-\frac{1}{2}$  all give the same probability,  $\frac{1}{4}$ .

**When amplitudes add, the relative phase comes back.** In quantum mechanics amplitudes, not probabilities, are what add. Expand the modulus squared of a sum using  $|w|^2 = ww^*$  and the conjugate rules:

$$|z_1 + z_2|^2 = (z_1 + z_2)(z_1^* + z_2^*) = |z_1|^2 + |z_2|^2 + z_1^*z_2 + z_1z_2^*.$$

The last two terms are conjugates of each other,  $z_1z_2^* = (z_1^*z_2)^*$ , and a number plus its conjugate is twice its real part. So

$$|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2\operatorname{Re}(z_1^*z_2).$$

The extra piece is the **cross term** (or interference term). With  $z_1 = r_1e^{i\theta_1}$  and  $z_2 = r_2e^{i\theta_2}$  it is  $2r_1r_2\cos(\theta_2 - \theta_1)$ . It depends on the *difference* of the phases, called the **relative phase**, and it can be positive (the two contributions reinforce), negative (they cancel) or zero.

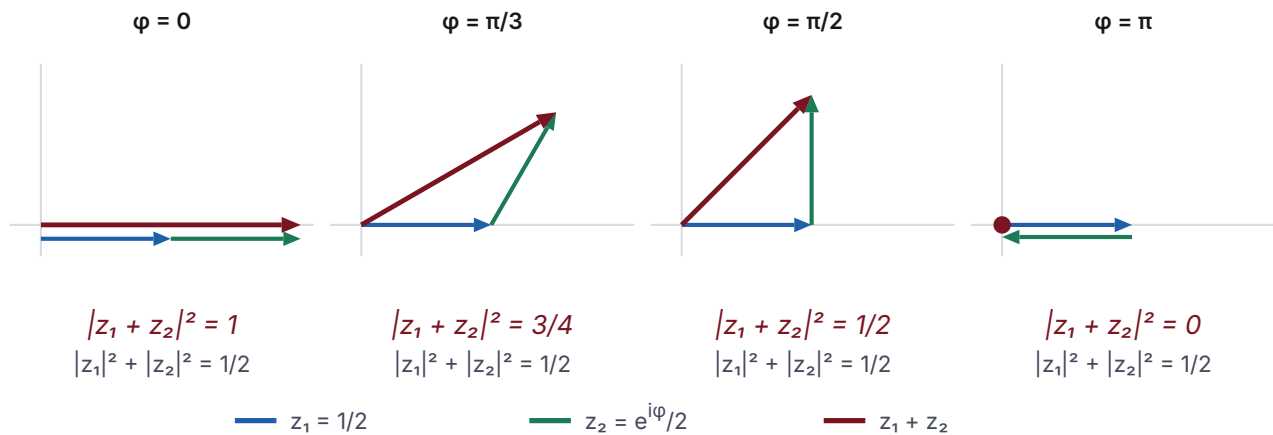
### WORKED EXAMPLE · TWO AMPLITUDES ADDING

Let  $z_1 = \frac{1}{2}$  and  $z_2 = \frac{1}{2}e^{i\pi/3} = \frac{1}{4} + \frac{\sqrt{3}}{4}i$ .

Separately,  $|z_1|^2 + |z_2|^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ .

Together,  $z_1 + z_2 = \frac{3}{4} + \frac{\sqrt{3}}{4}i$ , so  $|z_1 + z_2|^2 = \frac{9}{16} + \frac{3}{16} = \frac{3}{4}$ .

The cross term accounts for the difference:  $2 \operatorname{Re}(z_1^* z_2) = 2 \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{4}$ , and  $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ .



**Figure 1.7** The same two amplitude sizes, four relative phases. The sum of the separate squares is always  $\frac{1}{2}$ , but  $|z_1 + z_2|^2$  runs from 1 (the arrows line up) down to 0 (they point in opposite directions and cancel). That range is interference, and it is set entirely by the relative phase.

**A phase shared by everything changes nothing.** Multiply both amplitudes by the same phase factor  $e^{i\gamma}$ . Every modulus is unchanged, and every relative phase is unchanged because  $\gamma$  cancels in  $(\theta_2 + \gamma) - (\theta_1 + \gamma)$ . So no probability can change. A phase applied to the whole state is called a **global phase**, and Unit 3 will show that it has no physical effect at all, while a relative phase can change every measurement outcome. You now have the mathematics to see why.

### NOTE · CHECK IT IN NUMPY

The worked example on two amplitudes, plus a global phase to show it changes nothing.

```

:::python
import numpy as np
z1 = 0.5
z2 = 0.5 * np.exp(1j * np.pi / 3)
print(abs(z1 + z2)**2)           # 0.7499999999999999
print(abs(z1)**2 + abs(z2)**2)   # 0.5
print(2 * np.real(np.conj(z1) * z2)) # 0.25000000000000006
g = np.exp(1j * 0.7)             # any global phase
print(abs(g*z1 + g*z2)**2)       # 0.7500000000000001, unchanged

```

# Summary of the rules

For  $z = a + bi = re^{i\theta}$  and  $w = c + di = se^{i\phi}$ :

| OPERATION       | CARTESIAN FORM                                    | POLAR FORM                       |
|-----------------|---------------------------------------------------|----------------------------------|
| Sum $z + w$     | $(a + c) + (b + d)i$                              | convert to Cartesian first       |
| Product $zw$    | expand, use $i^2 = -1$ : $(ac - bd) + (ad + bc)i$ | $rs e^{i(\theta+\phi)}$          |
| Quotient $z/w$  | $\frac{zw^*}{ w ^2}$                              | $\frac{r}{s} e^{i(\theta-\phi)}$ |
| Conjugate $z^*$ | $a - bi$                                          | $re^{-i\theta}$                  |
| Modulus $ z $   | $\sqrt{a^2 + b^2}$                                | $r$                              |
| Modulus squared | $zz^* = a^2 + b^2$                                | $r^2$                            |
| Power $z^n$     | expand (slow)                                     | $r^n e^{in\theta}$               |
| Argument        | angle of $(a, b)$ , quadrant checked              | $\theta$                         |
| Powers of $i$   | $i^n = i^{n \bmod 4}$                             | $i = e^{i\pi/2}$                 |

And the identities worth knowing without looking up:  $(z + w)^* = z^* + w^*$ ,  $(zw)^* = z^*w^*$ ,  $z + z^* = 2 \operatorname{Re}(z)$ ,  $|zw| = |z||w|$ ,  $|e^{i\theta}| = 1$ ,  $e^{i\pi} = -1$ ,  $\frac{1}{i} = -i$ , and  $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2 \operatorname{Re}(z_1^* z_2)$ .

# Self-assessment

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The questions run from routine to harder. Do them on paper first; use NumPy only to check. Full solutions follow the questions.

1. Give the real and imaginary parts of  $-3 + 7i$ , of  $5$  and of  $-2i$ . Which is real, and which is purely imaginary?
2. Plot  $2 + i$ ,  $-1 + 3i$ ,  $-2 - 2i$  and  $4i$  in the complex plane and name the quadrant or axis of each.
3. Compute  $(4 - 2i) + (-1 + 5i)$  and  $(4 - 2i) - (-1 + 5i)$ .
4. Compute  $(3 + 2i)(1 - i)$ .
5. Simplify  $i^{10}$ ,  $i^{33}$  and  $i^{-3}$ .
6. For  $z = 5 - 12i$ , find  $z^*$ ,  $zz^*$  and  $|z|$ .
7. Find real numbers  $a$  and  $b$  such that  $(a + 2i) + (3 - bi) = 5 + 7i$ .
8. Compute  $\frac{2 + i}{1 - i}$  in the form  $x + yi$ .
9. Let  $z = a + bi$ . Show that  $z + z^*$  is real and  $z - z^*$  is purely imaginary. Then show that  $\frac{1}{i} = -i$  by multiplying the top and bottom by  $i$  first.
10. Verify  $|zw| = |z| |w|$  for  $z = 3 + 4i$  and  $w = 1 + i$  by computing both sides.
11. Write in polar form  $re^{i\theta}$  with  $-\pi < \theta \leq \pi$ :  $-2 + 2i$ ,  $-3$ ,  $-4i$ ,  $\sqrt{3} - i$ .
12. Write in Cartesian form:  $2e^{i\pi/3}$ ,  $5e^{i\pi}$ ,  $e^{-i\pi/2}$ ,  $\sqrt{2}e^{i5\pi/4}$ .
13. A student says  $\arg(-1 - i) = \arctan\left(\frac{-1}{-1}\right) = \frac{\pi}{4}$ . What went wrong, and what is the correct principal argument?
14. Use polar form to compute  $(1 + \sqrt{3}i)^6$ .
15. Let  $z = 2e^{i\pi/5}$  and  $w = 3e^{i3\pi/10}$ . Find  $zw$  and  $z/w$  in polar form, and write  $zw$  in Cartesian form.
16. Which single complex number, used as a multiplier, turns every arrow  $45^\circ$  anticlockwise and doubles its length? Apply it to  $3 + i$ .
17. Let  $z_1 = \frac{1}{2}$  and  $z_2 = \frac{1}{2}e^{i\phi}$ . Show that  $|z_1 + z_2|^2 = \frac{1}{2}(1 + \cos \phi)$  and evaluate it for  $\phi = 0, \pi/2$  and  $\pi$ . What happens to the answer if both  $z_1$  and  $z_2$  are first multiplied by  $e^{i\gamma}$ ?
18. Find both complex numbers  $z$  with  $z^2 = i$ . Give them in polar and Cartesian form and check one of them by squaring.

1.  $-3 + 7i$ : real part  $-3$ , imaginary part  $7$ .  $5$ : real part  $5$ , imaginary part  $0$ , so it is real.  $-2i$ : real part  $0$ , imaginary part  $-2$ , so it is purely imaginary.
2.  $2 + i$  is in the first quadrant (right, up).  $-1 + 3i$  is in the second (left, up).  $-2 - 2i$  is in the third (left, down).  $4i$  lies on the positive imaginary axis.
3. Sum:  $(4 - 1) + (-2 + 5)i = 3 + 3i$ . Difference:  $(4 + 1) + (-2 - 5)i = 5 - 7i$ .
4.  $(3 + 2i)(1 - i) = 3 - 3i + 2i - 2i^2 = 3 - i + 2 = 5 - i$ .
5.  $10 = 4 \times 2 + 2$ , so  $i^{10} = i^2 = -1$ .  $33 = 4 \times 8 + 1$ , so  $i^{33} = i$ . For  $i^{-3}$ , multiply by  $i^4 = 1$ :  $i^{-3} = i^{-3}i^4 = i$ .
6.  $z^* = 5 + 12i$ .  $zz^* = 25 + 144 = 169$ .  $|z| = \sqrt{169} = 13$ .
7. The left side is  $(a + 3) + (2 - b)i$ . Matching real parts,  $a + 3 = 5$ , so  $a = 2$ . Matching imaginary parts,  $2 - b = 7$ , so  $b = -5$ .
8. Multiply top and bottom by  $1 + i$ ; the bottom becomes  $(1 - i)(1 + i) = 2$ . The top is  $(2 + i)(1 + i) = 2 + 2i + i + i^2 = 1 + 3i$ . So the answer is  $\frac{1}{2} + \frac{3}{2}i$ .
9.  $z + z^* = (a + bi) + (a - bi) = 2a$ , which is real.  $z - z^* = (a + bi) - (a - bi) = 2bi$ , which is purely imaginary. For the second part,  $\frac{1}{i} = \frac{i}{i \cdot i} = \frac{i}{-1} = -i$ . Check:  $i \cdot (-i) = -i^2 = 1$ .
10.  $|z| = 5$  and  $|w| = \sqrt{2}$ , so  $|z||w| = 5\sqrt{2}$ . Also  $zw = (3 + 4i)(1 + i) = 3 + 3i + 4i + 4i^2 = -1 + 7i$ , and  $|zw| = \sqrt{1 + 49} = \sqrt{50} = 5\sqrt{2}$ . The two sides agree.
11.  $-2 + 2i$ :  $r = \sqrt{8} = 2\sqrt{2}$ , second quadrant, reference angle  $\pi/4$ , so  $\theta = 3\pi/4$  and  $z = 2\sqrt{2}e^{i3\pi/4}$ .  $-3$ : on the negative real axis,  $r = 3$ ,  $\theta = \pi$ , so  $z = 3e^{i\pi}$ .  $-4i$ : on the negative imaginary axis,  $r = 4$ ,  $\theta = -\pi/2$ , so  $z = 4e^{-i\pi/2}$ . (The angle  $3\pi/2$  is the same direction but lies outside the principal range.)  $\sqrt{3} - i$ :  $r = 2$ , fourth quadrant, reference angle  $\arctan(1/\sqrt{3}) = \pi/6$ , so  $\theta = -\pi/6$  and  $z = 2e^{-i\pi/6}$ .
12.  $2e^{i\pi/3} = 2(\frac{1}{2} + \frac{\sqrt{3}}{2}i) = 1 + \sqrt{3}i$ .  $5e^{i\pi} = 5(-1 + 0i) = -5$ .  $e^{-i\pi/2} = \cos(-\frac{\pi}{2}) + i\sin(-\frac{\pi}{2}) = -i$ .  $\sqrt{2}e^{i5\pi/4} = \sqrt{2}(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i) = -1 - i$ .
13. The ratio  $\frac{-1}{-1} = 1$  has lost both minus signs, and  $\arctan$  only returns angles in the right half of the plane. The point  $-1 - i$  is in the third quadrant, pointing directly away from  $1 + i$ . The reference angle is  $\pi/4$ , and in the third quadrant the principal argument is  $-\pi + \pi/4 = -3\pi/4$  (equivalently  $5\pi/4$ , outside the principal range).
14.  $1 + \sqrt{3}i$  has  $r = 2$  and  $\theta = \pi/3$ . So  $(1 + \sqrt{3}i)^6 = 2^6e^{i6\pi/3} = 64e^{i2\pi} = 64$ .
15.  $zw = 6e^{i(\pi/5 + 3\pi/10)} = 6e^{i5\pi/10} = 6e^{i\pi/2}$ , which in Cartesian form is  $6i$ .  $z/w = \frac{2}{3}e^{i(\pi/5 - 3\pi/10)} = \frac{2}{3}e^{-i\pi/10}$ .
16. The multiplier has modulus  $2$  and argument  $\pi/4$ :  $2e^{i\pi/4} = 2(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i) = \sqrt{2} + \sqrt{2}i$ . Applied to  $3 + i$ :  $(3 + i)(\sqrt{2} + \sqrt{2}i) = 3\sqrt{2} + 3\sqrt{2}i + \sqrt{2}i + \sqrt{2}i^2 = 2\sqrt{2} + 4\sqrt{2}i$ . Check:  $|3 + i| = \sqrt{10}$  and  $|2\sqrt{2} + 4\sqrt{2}i| = \sqrt{8 + 32} = \sqrt{40} = 2\sqrt{10}$ , twice as long.
17.  $|z_1|^2 = |z_2|^2 = \frac{1}{4}$  and  $z_1^*z_2 = \frac{1}{2} \cdot \frac{1}{2}e^{i\phi} = \frac{1}{4}(\cos \phi + i\sin \phi)$ , so  $2\operatorname{Re}(z_1^*z_2) = \frac{1}{2}\cos \phi$ . Adding,  $|z_1 + z_2|^2 = \frac{1}{4} + \frac{1}{4} + \frac{1}{2}\cos \phi = \frac{1}{2}(1 + \cos \phi)$ . At  $\phi = 0$  this is  $1$ , at  $\phi = \pi/2$  it is  $\frac{1}{2}$  and at  $\phi = \pi$  it is  $0$ : complete cancellation. Multiplying both by  $e^{i\gamma}$  changes nothing. Each modulus is unchanged, and  $(e^{i\gamma}z_1)^*(e^{i\gamma}z_2) = e^{-i\gamma}e^{i\gamma}z_1^*z_2 = z_1^*z_2$ , so the cross term is unchanged as well.
18. Write  $z = re^{i\theta}$ . Then  $z^2 = r^2e^{i2\theta}$  must equal  $i = 1 \cdot e^{i\pi/2}$ , so  $r = 1$  and  $2\theta = \pi/2$  or  $2\theta = \pi/2 + 2\pi$ . That gives  $\theta = \pi/4$  or  $\theta = 5\pi/4$ . The two roots are  $e^{i\pi/4} = \frac{1}{\sqrt{2}}(1 + i)$  and  $e^{i5\pi/4} = -\frac{1}{\sqrt{2}}(1 + i)$ . Check the first:  $(\frac{1}{\sqrt{2}}(1 + i))^2 = \frac{1}{2}(1 + 2i + i^2) = \frac{1}{2} \cdot 2i = i$ .

## Where to go next

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Unit 2 builds lists and grids out of complex numbers: vectors and matrices with complex entries. Every operation there reduces to the arithmetic of this unit. The conjugate becomes the conjugate transpose, and  $|z|^2 = zz^*$  becomes the norm of a vector, which is how a quantum state is made to have total probability 1. If the conjugate and the modulus feel slow, practise them now; they will appear on nearly every page from here on.

**Part A of the class assignment, The Qubit Workbook**, covers this unit and is due at the end of week 4. Work through the self-assessment above before you start it: every Part A question uses a skill practised here.